

1. DATA INPUTS



Yield Trial Data
K+ obs, georeferenced



Daily Rainfall
[CHIRPSV3.0]



Market Prices

[Maize, Fertilizer-Monthly]



Soil Properties

[SoilGrids, AfSIS]



Vegetation Indices

[NDVI/EVI Time Series]



Spatial Covariates

[Elevation, Access, SPAM]

2. PRE-PROCESSING [IN]

2. PRE-PROCESSING & INTEGRATION

Cleaning,
Georeferencing,
5km Grid,
Alignment,
Covariate Extraction

3. PREDICTIVE MODELLING

[Parallel Streams]

Market
data &
Covariates
[IN]

3a. Price Modelling

Trial data &
Covariates
[IN]

3b. Yield Response Modelling

Clean
Gridde
data
[OUT]

3c. Crop failer Probability Modelling

NDVI
Anomalies
&
Covariates
[IN]

4. UNCERTAINTY & DEPENDENCY

[$i=10,000$ iterations]

Predicted
Price
Surfaces
[OUT]

Historical Rain & Prices [IN]

Yield
response
Surfaces
[OUT]

**SPATIAL
COPULA**
[Joint Rainfall-Price]
[OUT]

Simulated
Joint
Scenarios
[Drawa]
[IN]

Yield
response
Surfaces
[IN]

CFP
Surfaces
[IN]

5. MONTE CARLO SIMULATION ENGINE

1. Yield Computation

$$\begin{aligned} Y_{NPK}^s &= \hat{f}(\text{soil}, \text{rain}^{(s)}, N, P, K) \\ Y_0^s &= \hat{f}(\text{soil}, \text{rain}^{(s)}, 0, 0, 0) \end{aligned}$$

2. Yield Gain

$$G^S = Y_{NPK}^S - Y_0^S$$

3. Returns

$$R^s = G^s \times Price^s$$

4. Crop-Loss Adjustment

$$\begin{aligned} \pi_{soft}^s &= (1 - p_{loss}^s) x R^s - C_{fert} \\ \pi_{hard}^s &= \begin{cases} -C_{fert} & \text{if crop fails} \\ R^s - C_{fert} & \text{Otherwise} \end{cases} \end{aligned}$$

Simulated
profit
Distribution
[OUT]

6. POST-SIMULATION OUTPUTS & DECISION SUPPORT

Statistical Metrics

Mean Profit, SD, VaR,
Cvar [Tail Risk]

Spatial Visualization

Figure 1 consists of four maps of Africa, each showing a different variable. The maps are arranged in a 2x2 grid. The top-left map shows the 'Probability of negative profit (P[Profit < 0])' with a color scale from 0 (blue) to 1 (red). The top-right map shows 'Crofting/Overcrowding Risk' with a color scale from 0 (blue) to 1 (red). The bottom-left map shows 'GDP per capita (USD)' with a color scale from 0 (blue) to 1 (red). The bottom-right map shows the 'Negative price ratio (P[Loss] / expected profit)' with a color scale from 0 (blue) to 1 (red). A legend in the top-right map identifies the symbols: a black dot for 'Non-irrigated cropland', a blue line for 'Major roads', and a red line for 'International borders'. The maps are labeled with latitude and longitude coordinates.

Decision Tools

Policy & Investment Insights